EXERCISE 7 – INTERPRETATION FOR THE DATASET INTRODUCED IN THE PREVIOUS SLIDE:

1. Compute the normalized information gain

$$\frac{I(X,Y_i)}{H(X)}$$

for each feature.

2. Identify the most informative and the least informative feature about the target X.

1. Dataset Overview & Target Definition

Using the 10-row sample from the **Titanic Passenger Survival Dataset**:

- **Target Variable (X - Survival):**
  - x_1: Survived ($P(X = x_1) = 0.40$)
  - x_2: Did Not Survive ($P(X = x_2) = 0.60$)
  - **Target Entropy:** $H(X) = -(0.4 \log_2 0.4 + 0.6 \log_2 0.6) \approx \mathbf{0.9710}$ bits
- **Features (Y_i):**
  - Y_1 **(Passenger Class):** Class 1, Class 2, Class 3
  - Y_2 **(Sex):** Female, Male
  - Y_3 **(Port of Embarkation):** Cherbourg, Southampton

2. Mutual Information and Normalized Information Gain Calculations

The normalized information gain (also known as the **Uncertainty Coefficient** or **Information Gain Ratio**) measures the fraction of entropy in the target X that is explained by observing feature Y_i:

$$\text{Normalized Info Gain} = \frac{I(X;Y_i)}{H(X)}$$

Feature (Yi)	Mutual Information I(X;Yi)	Target Entropy H(X)	Normalized Information Gain H(X)I(X;Yi)
Y_1 **(Passenger Class)**	0.2142 bits	0.9710 bits	0.2206 (22.06%)
Y_2 **(Sex)**	0.9710 bits	0.9710 bits	1.0000 (100.0%)
Y_3 **(Port of Embarkation)**	0.0610 bits	0.9710 bits	0.0628 (6.28%)

3. Identification of Most and Least Informative Features

- **Most Informative Feature:**
  - Y_2 **(Sex)** is the most informative feature, achieving a normalized information gain of 1.0000 **(**100%**)**.
  - *Interpretation:* In this specific sample, knowing a passenger's sex completely eliminates uncertainty about their survival ($H(X|Y_2) = 0$). All females survived and all males did not in this subset, providing maximum predictive power.
- **Least Informative Feature:**
  - Y_3 **(Port of Embarkation)** is the least informative feature, yielding a normalized information gain of only 0.0628 **(**6.28%**)**.
  - *Interpretation:* Knowing where a passenger embarked provides very little reduction in uncertainty regarding whether they survived, making it a weak predictor for the target variable X.